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Issue:	1
Date:	26 August 2008
Compliance date:	26 August 2008

Level 3

Procedure and competence requirements for persons undertaking works in the vicinity of high voltage cables

Endorsement and Authorisation

Endorsed by:



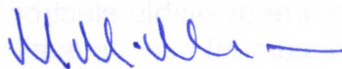
Philip Frazer, Working Group Chair

Authorised by:



Kevin Lydford, Steering Group Chair

Accepted for issue by:



Mick McManus, National Standards Manager

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Issue record

Issue	Date	Comments
1	26 August 2008	Phase 2A / Engineering reorganisation responsibility change

Compliance

This Network Rail standard is mandatory and shall be complied with by Network Rail and its contractors if applicable from 26 August 2008.

When this standard is implemented, it is permissible for all projects that have formally completed GRIP Stage 4 to continue to comply with the Issue of any relevant Network Rail Standards current when GRIP Stage 4 was reached and not to comply with requirements contained herein, unless the designated Standard Owner has stipulated otherwise in the accompanying Briefing Note.

Reference documentation

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1 Purpose

To confirm works undertaken in the vicinity of High Voltage cables are conducted in a manner which minimises risk to persons, cables and the environment.

2 Scope

This procedure applies to all persons requiring to undertake work in the vicinity of the South East area HV networks both those cables controlled from the Electrical Control Room and the independent uncontrolled systems. The procedure applies to all such Network Rail owned HV cables wherever located or positioned.

This procedure excludes all works which involve access to the cables themselves, associated joints, terminations or other distribution equipment or switchgear.

3 Roles and responsibilities

The Principal Maintenance Support Engineer (M&E) is responsible for establishing the process and competence standards of persons undertaking work in the vicinity of Network Rail's HV cables.

4 Definitions

For the purposes of this procedure the following definitions apply:-

HV Cable - any cable operating at over 1000volts a.c (and associated pilot/supervisory cable(s)) forming part of the South East areas High Voltage Distribution networks (including separate networks not controlled from an Electrical Control Room).

Principal Maintenance Support Engineer (M&E) PMSE (M&E) or his representative.

Competent Engineer - the electrical engineer who is delegated in writing by the PMSE (M&E) to appoint Assessors of work to be undertaken in the vicinity of HV cables.

Assessor - person appointed in writing by the Competent Engineer to assess the risks presented by works to be undertaken in the vicinity of HV cables.

Competent Person - person appointed in writing by the Assessor to supervise protection measures for HV cables (including the on site execution of a cable switch out).

Responsible manager – Network Rail Contract Manager or Project Manager responsible for any works to be undertaken in the vicinity of HV cables.

Electrical Control Room (ECR) - location controlling a geographic area of the HV distribution network shown on Network Rail drawing SR/F/267.

Electrical Control Operator (ECO) - Person having day to day control & operation of the South East areas main High Voltage Distribution network including cables (as shown on Network Rail drawing SR/F/267) within a geographic area.

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HV Co-Ordinator (HVCO) - Person responsible for the programming of preplanned outages of the South East areas main HV Distribution network including cables (as shown on Network Rail drawing SR/F/267).

Contractor - persons undertaking work in the vicinity of HV cables.

Cable switch out - A cable that has been de-energised by the opening of circuit breakers or similar devices (e.g. fuse switches).

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5 Main clauses

5.1 Responsible manager

5.1.1 The Responsible manager for any excavation, vehicular movement, construction tasks or other activity which could damage a cable or the route containing the cable undertaken within the vicinity (1 metre) of such HV cables shall adhere to this procedure in order to control the risk of injury to persons and unwanted effects on the distribution network and the operational railway.

5.1.2 The Responsible manager shall confirm that persons undertaking such work are audited for compliance with this procedure.

5.1.3 Where works are to be undertaken in the vicinity of HV cables the Responsible manager shall confirm a Competent Engineer appoints Assessors to evaluate the risks associated with the tasks to be carried out in the vicinity of the cables.

5.2 Competent Engineers

5.2.1 The PMSE (M&E) shall delegate in writing the authority to Competent Engineers to evaluate the competence of, and appoint Assessors.

5.2.2 Within the area of his/her professional responsibility the Competent Engineer shall formally accept accountability for ensuring such measures as are necessary for the protection of HV cables and persons in the vicinity are provided and enforced.

5.2.3 The Competent Engineer shall, as appropriate, accept method statements where measures to control the risks to HV cables are proposed.

5.3 Assessors

5.3.1 The Assessor shall on each occasion determine the risks associated with any works in the vicinity of HV cables and measures for eliminating or controlling such risks including, where necessary, the switch out of the cables.

5.3.2 The Assessor shall on each occasion where protective measures are required in conjunction with works in the vicinity of HV cables appoint a Competent Person(s) for the site works.

5.3.3 Assessors shall be appointed for the duration of a Contract up to a maximum of two years.

5.4 Competent Persons

5.4.1 The Competent Person shall be responsible for ensuring all protective measures specified by the Assessor are in place before work commences and where necessary execute a switch out of the affected cables in accordance with this procedure.

5.4.2 Competent Persons shall be appointed for the duration of a Contract up to a maximum of two years.

6 Procedure

6.1 Assessment

6.1.1 When work is required to take place in the vicinity of an HV cable the Responsible manager shall confirm an Assessor is appointed who will decide on the

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basis of a risk assessment what measures to protect the HV cables/ persons are required. The assessment shall consider the task to be undertaken, the proximity of the cables and the precautions which can be employed to avoid damage to the cables. For repetitive tasks generic assessments may be employed as part of the initial evaluation. However, all specific site conditions must additionally be considered and included in overall assessment.

An individual site survey for the presence of other services shall form part of all assessments where any ground penetration is to take place.

The details of all assessments shall be formally recorded.

6.1.2 Having determined what protective measures are to be applied the Assessor shall confirm these are incorporated in the Contractors method statement for the task. Where the result of the assessment requires this shall include switching out of the HV cables.

6.1.3 The Assessor shall as appropriate appoint a Competent person to confirm the protective measures identified are applied at the site of the work, and where necessary, execute a switch out of the HV cable(s).

6.1.4 Where the HV cable(s) concerned are controlled from an Electrical Control Room the Assessor shall advise, in writing, the appropriate controlling location(s) the name, company and identifying reference number of any Competent Person he/she appoints for the works. The details of the Competent Person(s) to be employed on a given task shall, where relevant, be included on documentation provided to the ECO(s) in connection with that task.

A copy of advice of the appointment of any Competent Person(s) shall be provided to the PMSE (M&E).

6.2 Execution of protective measures

6.2.1 The Competent Person shall confirm the protective measures identified from the assessment (including where necessary a switch out of the cables) are in place before allowing work to commence. The Competent Person shall brief the persons undertaking the work as to the extent and any limitations of the protective measures used (e.g. presence of live cables or other hazards).

6.2.2 Should any of the protective measures become ineffective during the execution of the task work shall cease and protection either be reinstated or work suspended for further assessment. Should damage occur to any of the cables or route the task shall be at once suspended and the Assessor informed. Where Electrical Control(s) have control of the affected cables the duty ECO shall be immediately informed and such protective measures as the ECO directs undertaken by the Competent Person.

The Competent Person(s) shall remain at site for the duration of any physical works.

6.2.3 If a switch out of HV cables controlled by an Electrical Control is required it shall be arranged with the HV Co-Ordinator at Waterloo General Offices.

When the switch out has been agreed the Assessor shall complete form HV2 (Appendix A) giving all the required information and send it in advance of the work to the ECO at the appropriate Electrical Control. The duty ECO shall check the details of the proposed switch out in accordance with Electrical Control Room Instructions NR/ECR/001.

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6.2.4 Where the affected HV cables are not controlled from an Electrical Control (eg. 11KV domestic supplies) the Assessor shall complete form HV2A, (Appendix B) and send it to the persons undertaking the switchout.

6.2.5 A copy of the completed HV2 or HV2A form (as appropriate) shall be in the possession of the Competent Person who is to request the switch out. The Competent Person shall also possess all available cable/route diagrams etc. relevant to the cables concerned and any available diagrams of other services present (surface or buried).

6.2.6 Where the cables concerned are controlled from an Electrical Control, upon execution of the switch-out, the Competent Person shall confirm the details contained on form HV2 with the duty ECO and request the cables identified on the form be switched out. The duty ECO shall advise the Competent Person when he/she has done so.

A means of communication shall be available to the Competent Person which enables contact with the ECO.

6.2.7 Where cables concerned are not controlled from an Electrical Control the Competent Person shall obtain endorsement of his form HV2A from the persons undertaking the switchout that the cable(s) have been switched out.

6.2.8 If the Competent Person is relieved by another Competent Person this shall be recorded on form HV2 (or HV2A) and where cables are controlled from an Electrical Control, the ECO shall be informed.

6.2.9 Upon completion or suspension of the works the Competent Person shall inspect the cables/route to confirm no damage has occurred. He/she shall inform the work party the cables are to be recharged and no further work must take place. When all persons and equipment are removed and the cable(s)/route are returned to the required condition, the Competent Person shall advise the ECO or persons undertaking the switch-out it is safe to recharge the cables. The ECO or persons undertaking the switch-out shall return the cable(s) to service using normal procedures.

7. Competence

7.1 Competent Engineer

The Competent Engineer shall normally be the Engineer identified in the Contractors Safety Case as having professional responsibility for all works in respect to High Voltage Distribution cables and equipment.

For Network Rail this shall include the PMSE (M&E), designated representative and the Infrastructure Investment Project Engineer(s).

The PMSE (M&E) may authorise other appropriately qualified and experienced engineers as Competent Engineers at his/her discretion.

7.2 Assessors

The following core competency requirements shall apply to all persons authorised as Assessors

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- a) shall have at least 18 months experience of electrical power supply or similar equipment;
- b) shall have had at least 6 months experience in the handling and hazards of high voltage
- c) cables and had training on the specific hazards and risks associated with oil filled cables;
- d) shall understand the hazards associated with electrical fires;
- e) shall have received specific training on the HV cables types concerned including safe working in close proximity to live cables and the limits of safe working;
- f) shall be trained in basic risk assessment techniques such to be able to evaluate tasks to be undertaken in the proximity of HV cables;
- g) shall understand the procedures for gaining access to electrical equipment and the need to keep the ECO or other person in charge of the system advised of any actions or movements whilst at the location concerned;
- h) shall understand the scope of the appointment and be in possession of a valid certificate of competence issued by the Competent Engineer;
- i) h) shall have received training in setting up safe systems of work and issuing safety documentation;
- j) shall have passed an examination to the required level of competency.

7.3 Competent Persons

The following core competency requirements shall apply to all persons authorised as Competent Persons:-

- a) shall be at least 18 years of age ;
- b) shall have received training in the avoidance of electrical hazards on electrical power supply or similar equipment ;
- c) shall have had experience in the handling and hazards of high voltage cables and had training on the specific hazards and risks associated with oil filled cables ;
- d) shall hold a current certificate in First Aid for Electrical Accidents ;
- e) shall possess good written and verbal communication skills in English to a level necessary to avoid danger to themselves and others;
- f) shall be medically fit to a standard equivalent to that required for the issue of a Personal Track Safety Certificate ;
- g) shall understand the hazards associated with electrical fires ;
- h) shall be in possession of a copy of and understand the appropriate Electrical Safety Rules where provided ;
- i) shall understand the procedures for gaining access to electrical equipment and the need to keep the ECO or other person in charge of the system advised of any actions or movements whilst at the location concerned;

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- j) shall understand the scope of the appointment and be in possession of a valid certificate of competence issued by the Assessor ;
- k) shall have passed an examination to the required level of experience.

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Appendix A Form HV2

Switch-out for Work Adjacent to High Voltage Cable **- Cables Controlled from Electrical Control**

Network Rail
Form HV2

Electrical Control Room Telephone Number Internal:..... BT:.....

Feeder No.(s)..... From..... To.....

Details of Work.....
.....

Limits of Work Area.....
.....

Work Assessed by..... Company.....

Planned Commencement of Switch-out Time..... Date.....
Planned Completion of Switch-out Time..... Date.....

Competent Person requesting Switch-out Name Name (Relief).....

Contractor..... Telephone No.....

Location at Site.....

I confirm that I clearly understand which cables are switched out and the limits of safe working permitted

Competent Person: Block
Capitals.....

Competent Person (Relief): Block
Capitals.....

Signature..... Signature (Relief).....
Time..... Date..... Time..... Date.....

Completion of Work

I confirm that I have inspected Feeder No. (s)..... and its route and confirm that all work being carried out adjacent to the cable(s) has ceased and all staff have been informed the cables are to be made live.

Signed (Competent Person).....

Date..... Time.....

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Appendix B Form HV2A

Switch-out for Work Adjacent to High Voltage Cable

Network Rail
Form HV2A

Feeder No.(s)..... From..... To.....
Details of Work.....

Limits of Work Area
.....

Work Assessed by..... Company.....

Planned Commencement of Switch-out Time..... Date.....
Planned Completion of Switch-out Time..... Date.....

Competent Person requesting Switch-out Name Name (Relief).....

Contractor..... Telephone No.....

Location at Site.....

I confirm that I clearly understand which cables are switched out:

Name..... Contractor.....

Location..... Telephone Number.....

I confirm that I clearly understand which cables are switched out and the limits of safe working permitted:-

Competent Person: Block Capitals.....

Competent Person (Relief): Block Capitals.....

Signature..... Signature (Relief).....

Time..... Date..... Time..... Date.....

Completion of Work

I confirm that I have inspected Feeder No. (s)..... and its route and confirm that all work being carried out adjacent to the cable(s) has ceased and all staff have been informed the cables are to be made live.

Signed (Competent Person).....

Date..... Time.....

Standards Briefing Note



Ref: NR/L3/ELP/22001	Issue: 1	Publication Date: 26/08/2008	Compliance Date: 26/08/2008
Title: Procedure and competence requirements for persons undertaking works in the vicinity of high voltage cables			
Standard Owner: Head of Electrification and Plant			
Non-Compliance rep (NRNC): Head of Electrification and Plant			

Purpose: To confirm works undertaken in the vicinity of High Voltage cables are conducted in a manner which minimises risk to persons, cables and the environment.	The following teams require awareness briefing: Executive Management Group ☐ Commercial Property ☐ Contracts and Procurement ☐ Strategic Change ☐ CTRL ☐ Engineering Asset Management ☐ Civil Engineering ☐ E&P Engineering ☒ Enhancements Engineering ☐ Ergonomics ☐ Future Railway Programme ☒ Ops Principles & Standards ☐ Rail Vehicle Engineering ☐ Railway Systems ☐ Telecoms Engineering ☒ Track Engineering ☒ Signal Engineering ☒ Finance ☐ Funding ☐ Govt & Corp Affairs ☐ Human Resources ☐ Information Management ☐ Infrastructure Investment Crossrail ☒ Track ☒ Programme Management ☒ Contracts & Procurement ☐ HSQE ☐ Sig. Power & Comms ☒ WCRM ☐ Construction ☒ FTN/GSM-R ☒ Thameslink ☒ Enhancements ☒ Infrastructure Maintenance Maintenance Areas ☒ Operational Property ☐ Overhead Condition ☐ Renewals ☐ Legal Services ☐ National Delivery Service ☐ Network Development ☐ Operations & Customer Services ☒ Planning & Regulation ☐ Safety and Compliance ☐ Westwood ☐				
Scope: This Work Instruction applies to all persons requiring to undertake work in the vicinity of the South East area HV networks both those cables controlled from the Electrical Control Room and the independent uncontrolled systems. The procedure applies to all such Network Rail owned HV cables wherever located or positioned.					
What's New/Changed: References to Railtrack and Region amended and references to Electrification & Plant Engineer and Project Delivery Electrification & Plant Design & Construction Engineers have been changed; Electrification & Plant Engineer = Principal Maintenance Support Engineer (M&E) Project Delivery Electrification & Plant Design & Construction Engineers = II Project Engineers					
Affected documents: <table><tr><td><i>Reference</i></td><td><i>Impact</i></td></tr><tr><td>Territory Local Instruction 8.4.25, Issue 2</td><td>To be replaced with NR/L3/ELP/22001, Issue 1</td></tr></table>	<i>Reference</i>	<i>Impact</i>	Territory Local Instruction 8.4.25, Issue 2	To be replaced with NR/L3/ELP/22001, Issue 1	
<i>Reference</i>	<i>Impact</i>				
Territory Local Instruction 8.4.25, Issue 2	To be replaced with NR/L3/ELP/22001, Issue 1				
Implementation requirements: <i>The following posts have specific responsibilities within this standard and shall receive technical briefing as part of the Implementation Programme:</i> Principal Maintenance Support Engineer (M&E) II Project Engineers					
For further information contact: Name: Philip Frazer Contact number: 085 78980 Email: philip.frazer@networkrail.co.uk					